hcs-workcell

Setting up the cell and running it

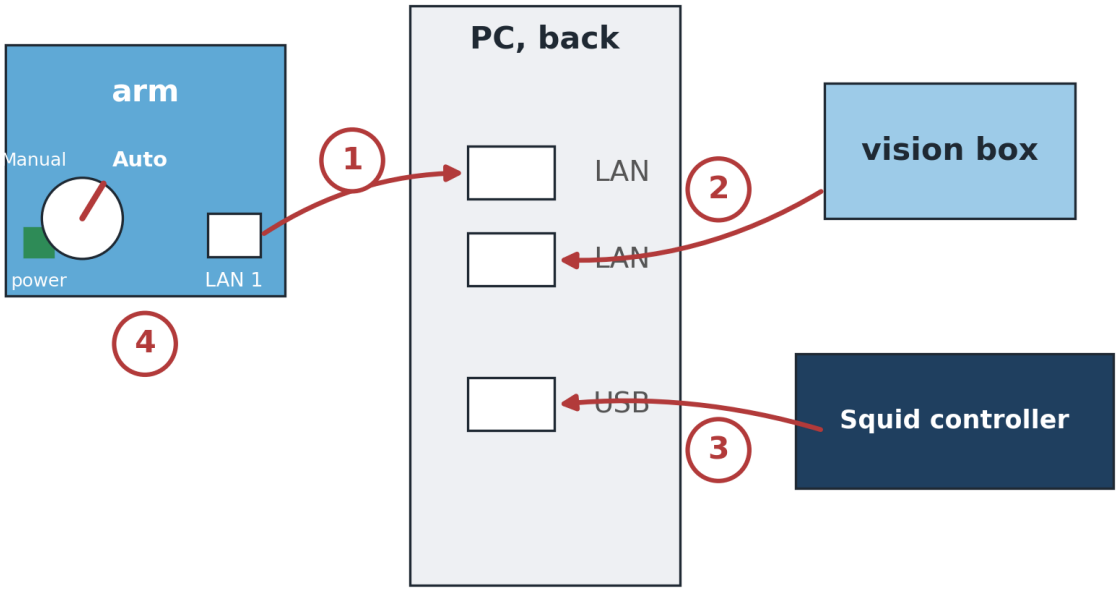
Two programs. setup: Setup, Teach, Test. run: Start, STOP.

0. Cables
1. Cameras, once
2. Setup
3. Teach
4. Test
5. Run
6. If something stops

STOP, on screen, at any time: the arm holds still.

0 Cables

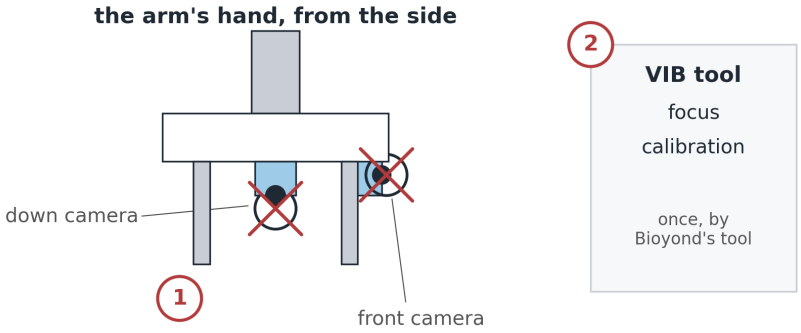
Plug in the three cables.
Switch the arm on. Key at Auto.



1 arm, port LAN 1, to the PC
2 vision box to the PC
3 Squid controller USB to the PC
4 arm power on, key at Auto

1 Cameras, once

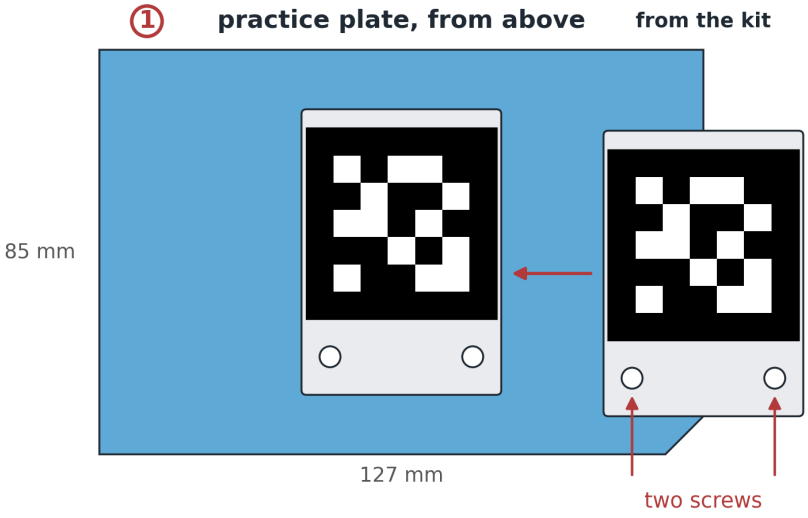
Focus and calibration are done once, with Bioyond's VIB tool. Do not touch the focus rings.
A camera moved or refocused means the VIB tool again, then Teach again.



1 the two focus rings: do not touch them
2 a camera moved or refocused: the VIB tool again, then this manual again

1.1 The marker tab

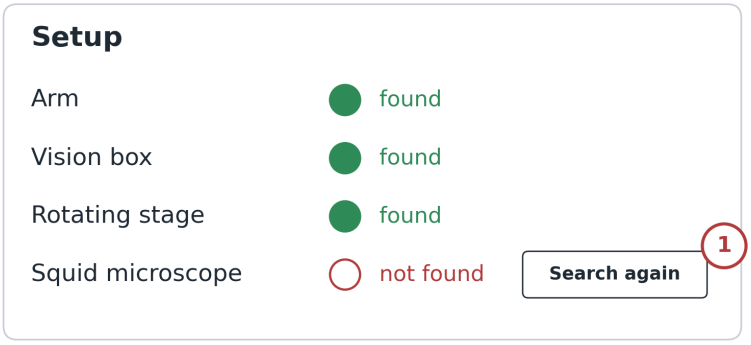
Screw the marker tab from the kit onto the practice plate.
The box reads the marker and works out where the plate is.



1 screw the tab onto the practice plate. The box reads the marker

2 Setup

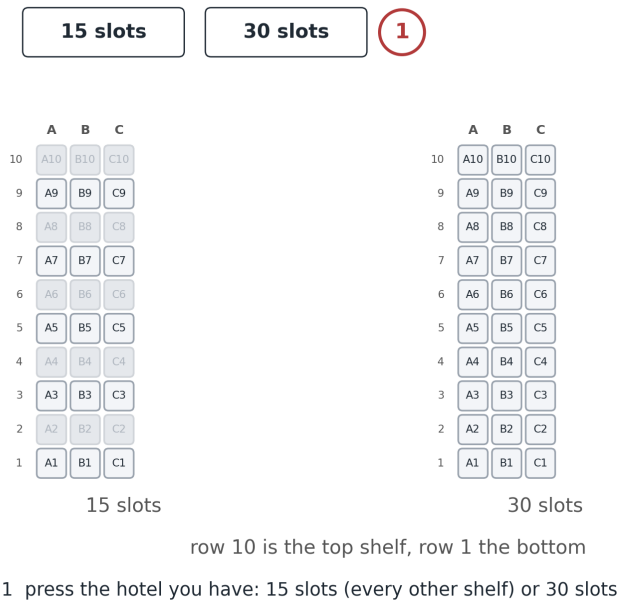
Open the setup program. Setup, Teach and Test live in it; run is the other program.
It finds the arm and the vision box by itself. Nothing to type. Every row green: go on.



1 a row not green: power, key at Auto, that device's cable, then Search again

2.1 The hotel

Press the hotel you have: 15 slots or 30 slots.



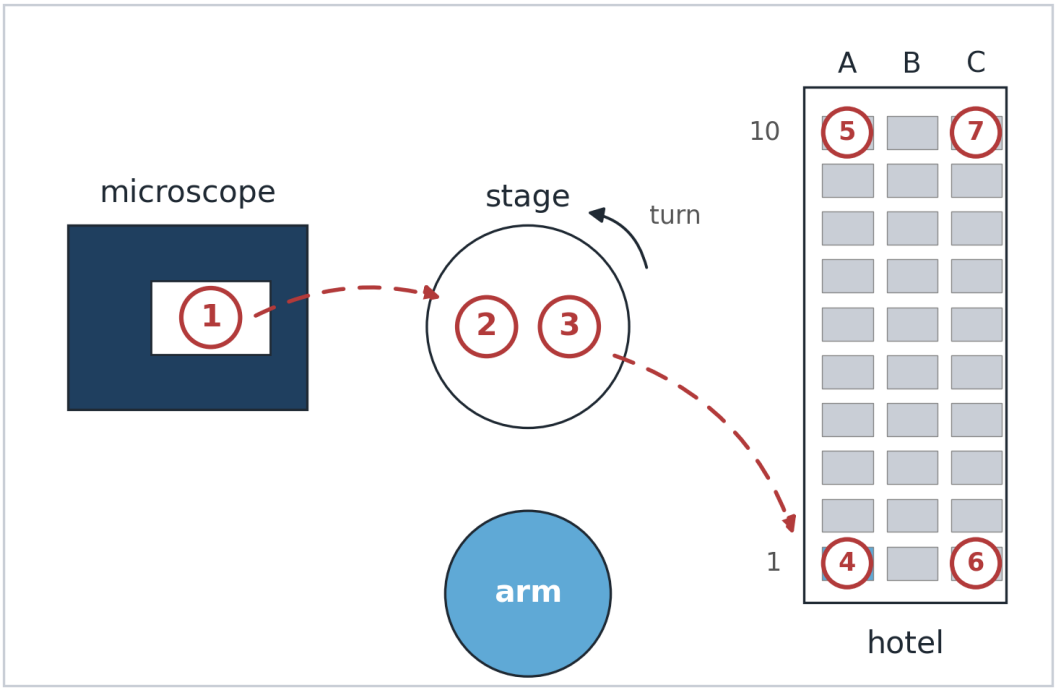
row 10 is the top shelf, row 1 the bottom

1 press the hotel you have: 15 slots (every other shelf) or 30 slots

3 Teach

One list. Put the practice plate where the row says, hover the hand over it so the marker is in view, press Record.
Go down the list. The way you drag the arm between two Records is the route it will use.

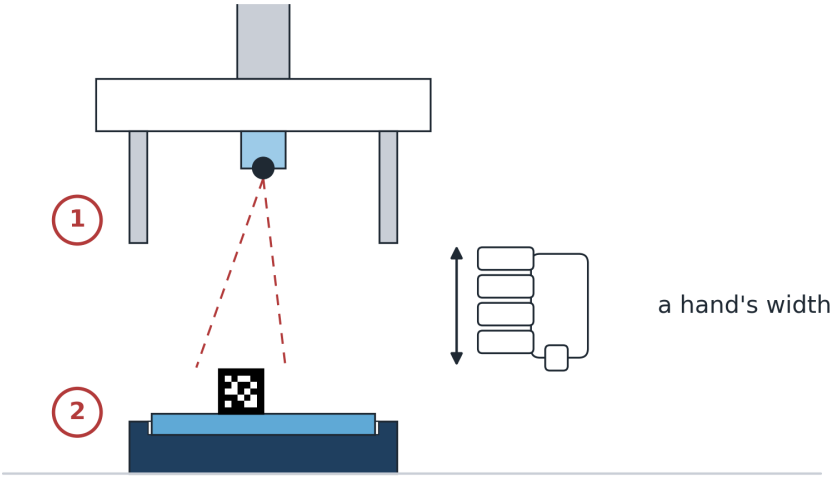
put the practice plate at each mark, hover the hand over it, press Record



1 the microscope holder, plate in the holder
2 stage at 0, plate seated, short edges in the fingers
3 stage at 90, plate seated, long edges in the fingers
4 hotel A1 5 hotel A10 6 hotel C1 7 hotel C10
dashed: the drag between two Records is the route the arm will use

3.1 Hover

A hand's width above the plate. The marker whole in the down camera's picture.

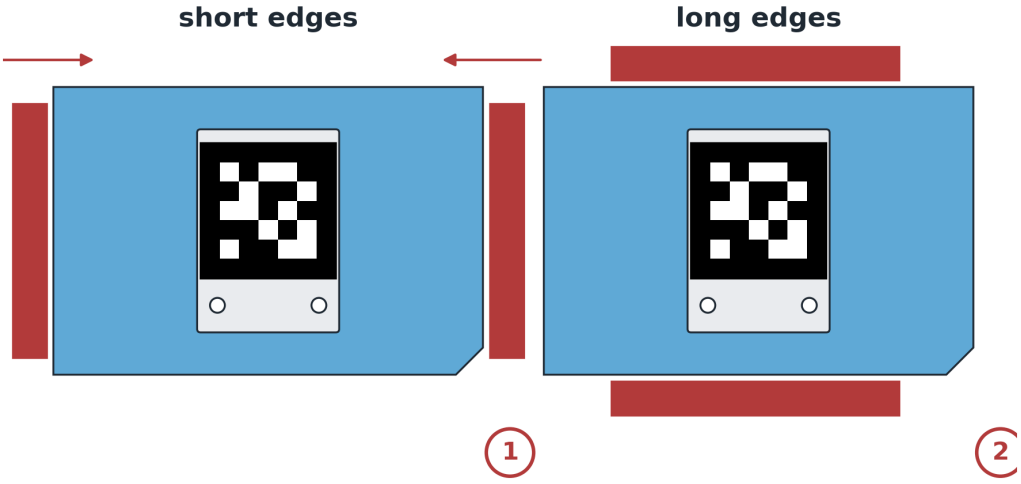


1 a hand's width up, the marker whole in the picture: press Record

2 the plate rests where it will rest in a run

3.2 Grip

Plate between the fingers. Press Measure with the short edges in the fingers, then again with the long edges.

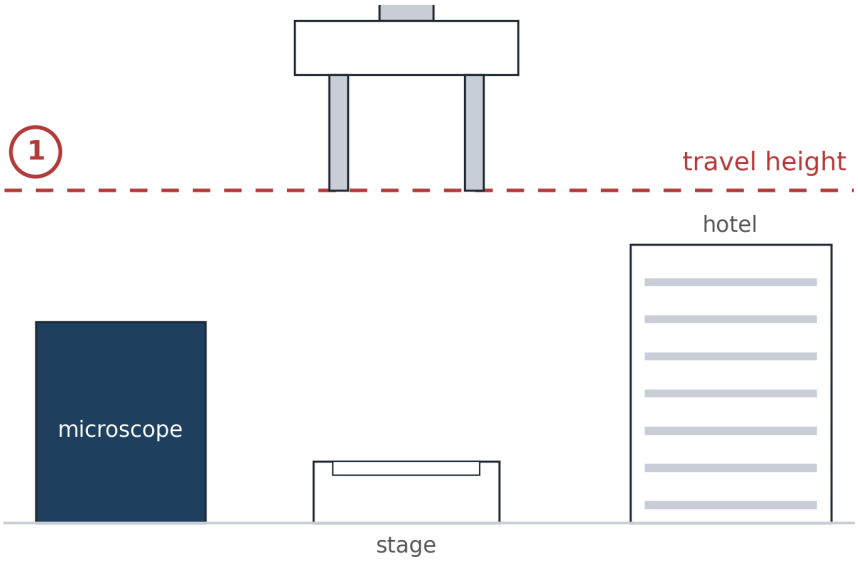


1 short edges, the two ends: the microscope holder and the stage at 0

2 long edges, the two sides: the stage at 90 and the hotel

3.3 Travel height

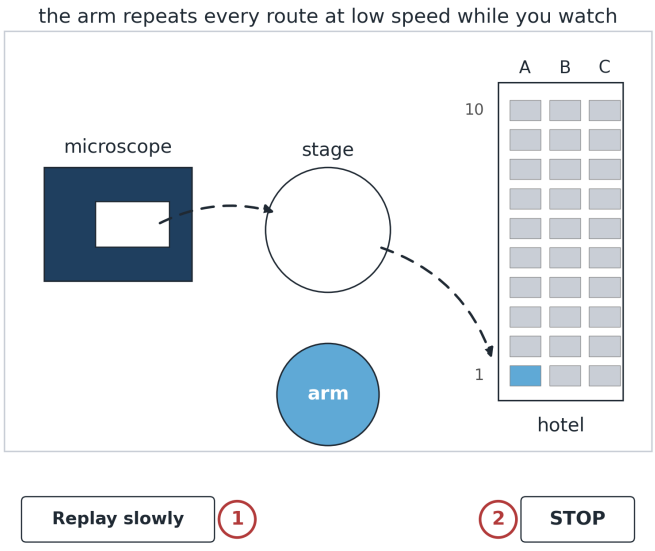
Lift the hand until it clears everything on the bench. Press Record.



1 lift the hand until it clears everything on the bench, press Record

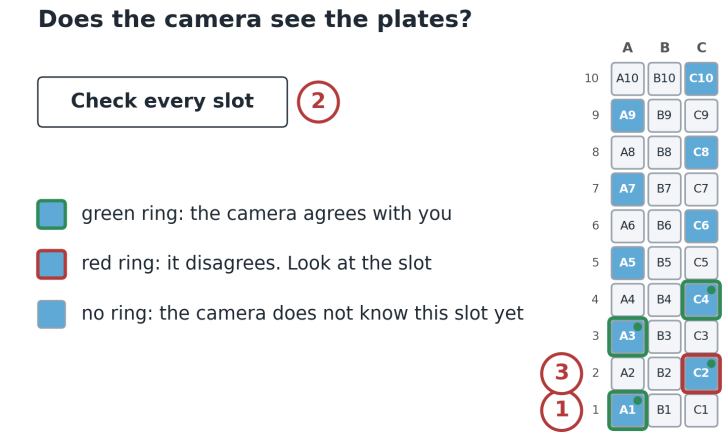
3.4 Replay slowly

Press Replay slowly. The arm repeats every route at low speed while you watch.
Done when no row in the list is red.



4 Test

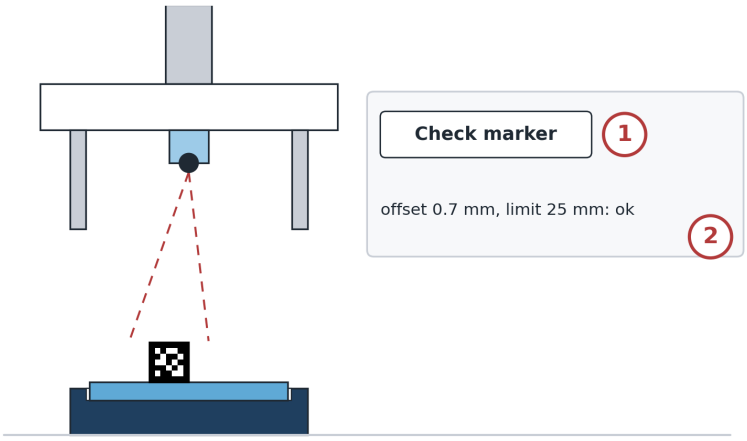
Click the slots that really hold a plate. Press Check every slot: the arm looks into each one.
A red ring: go and look at that slot.



1 click the slots that really hold a plate
2 the arm looks into each slot
3 a red ring: go and look at that slot

4.1 Marker

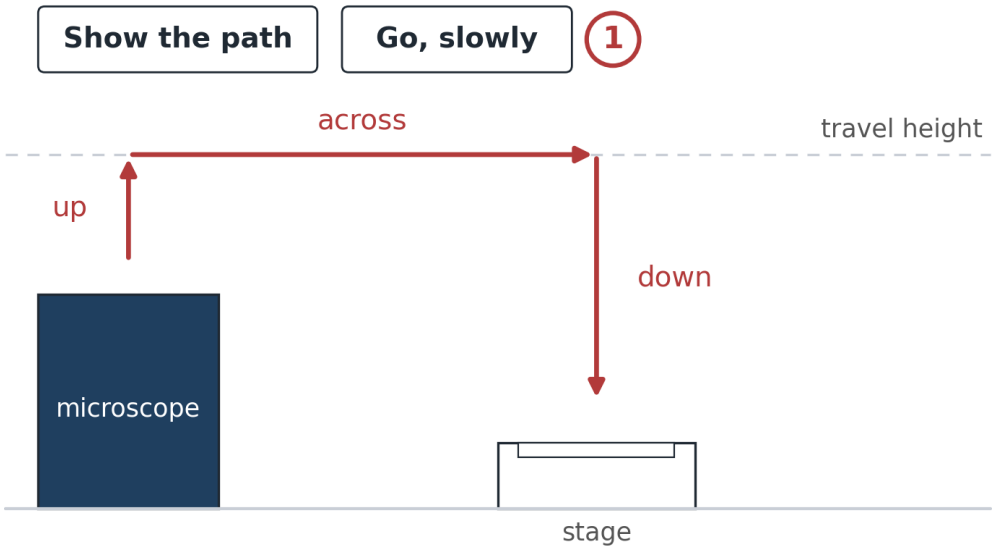
Pick a position, press Check marker. It says how far the position has moved.
Over the limit: teach that position again.



1 pick the position, press Check marker. The arm reads the marker
2 over the limit: teach that position again

4.2 The path

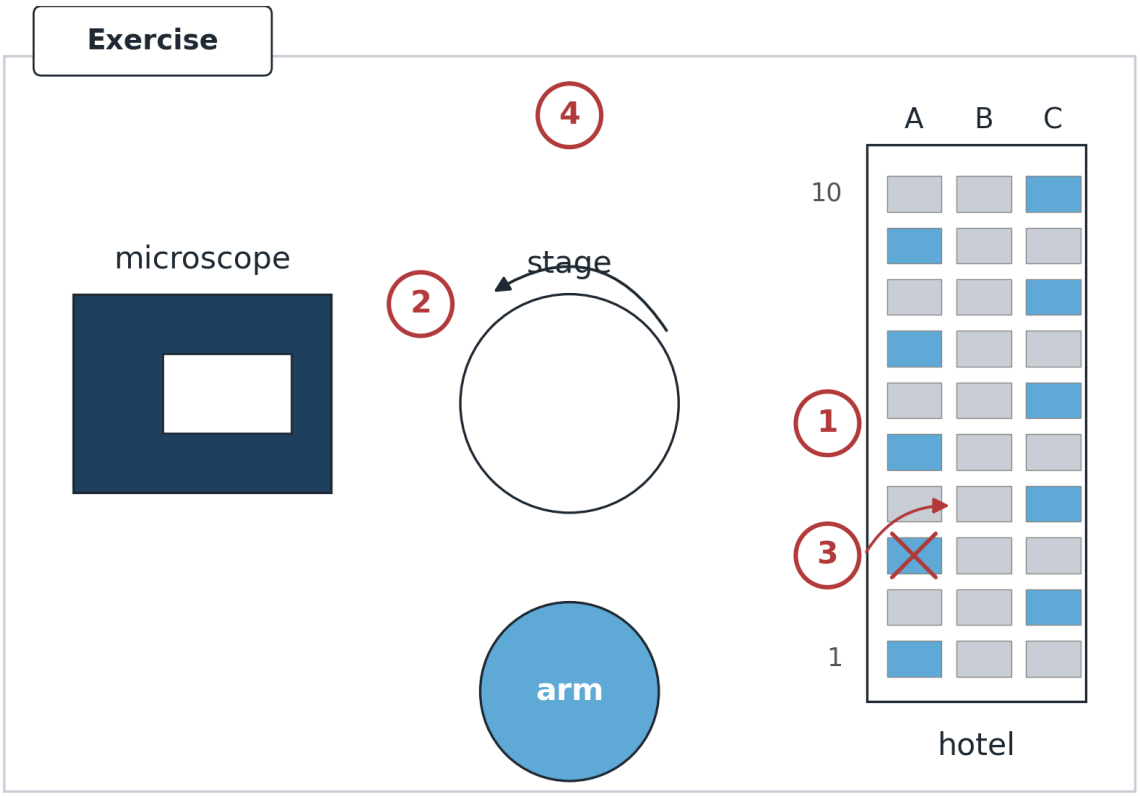
Pick from and to, press Show the path. It reads back the moves. Then Go, slowly, and watch.



1 pick from and to, press Show the path, then Go, slowly

4.3 Exercise

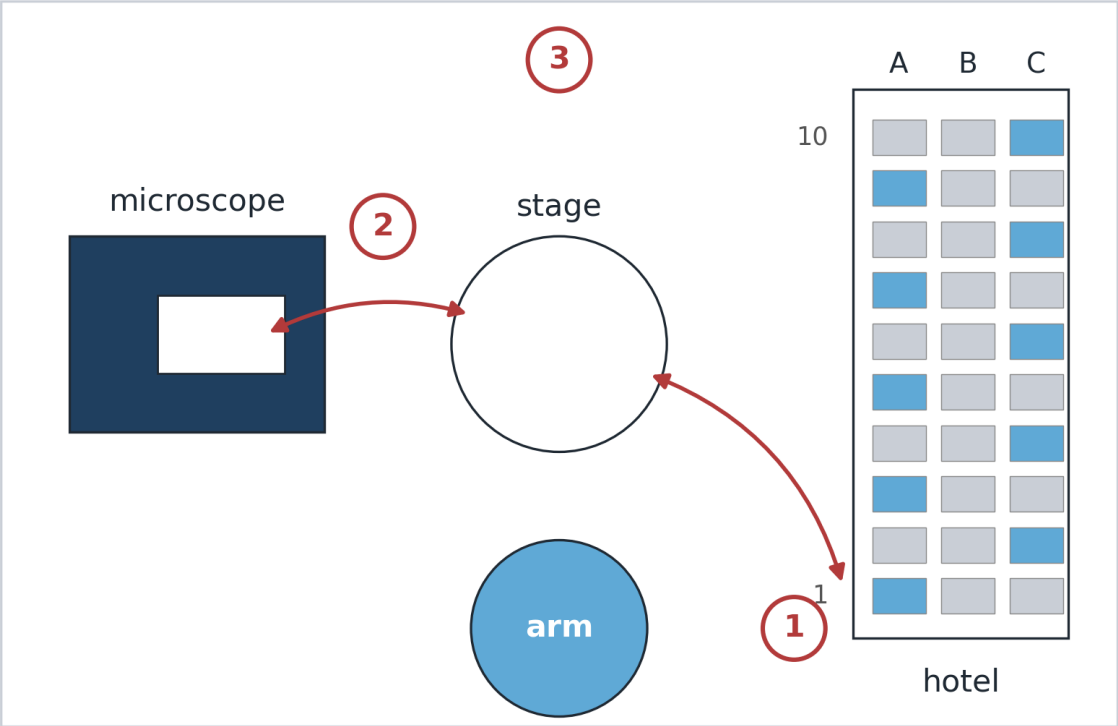
Press Exercise. On the simulator first, then on the arm.
The arm looks into every slot, turns the stage, orders a plate into a taken slot on purpose and backs out.



1 looks into every slot
2 turns the stage
3 orders a plate into a full slot on purpose, backs out, uses another slot
4 then runs all plates. On the simulator first, then on the arm

4.4 Exercise: every plate, in turn

Then it takes every plate to the microscope and back, one after the other.
Watch. STOP at any time.
Done when every plate is back in its own slot and the screen says finished.




```
A1  C2  A3  C4  A5  C6  A7  C8  A9  C10
```

1 one plate out of the hotel, over the stage, into the microscope holder
2 back out, over the stage, into its own slot. Then the next plate
3 the order: A1, C2, A3, ... columns alternate so the arm swings less

5 Run

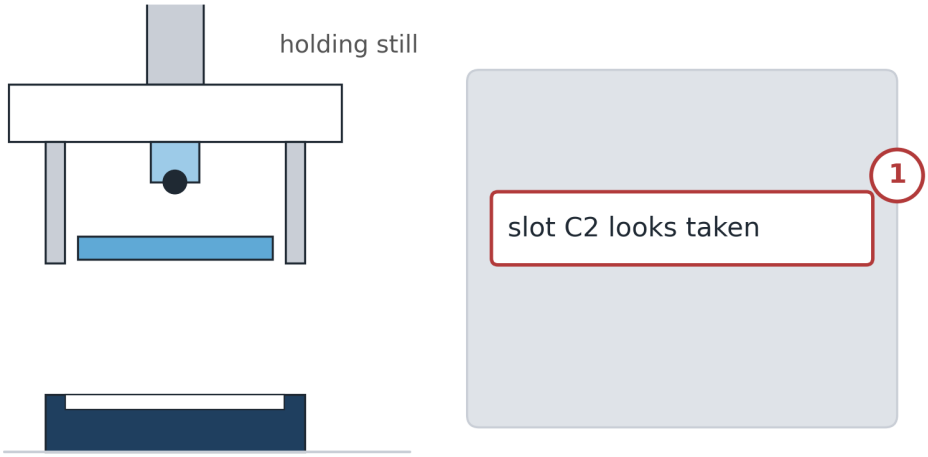
When the tests pass, open the run program and press Start.



1 the arm looks into every slot, shows what it found, then runs every plate in turn
2 STOP at any time. The arm holds still. Start again continues

6 If something stops

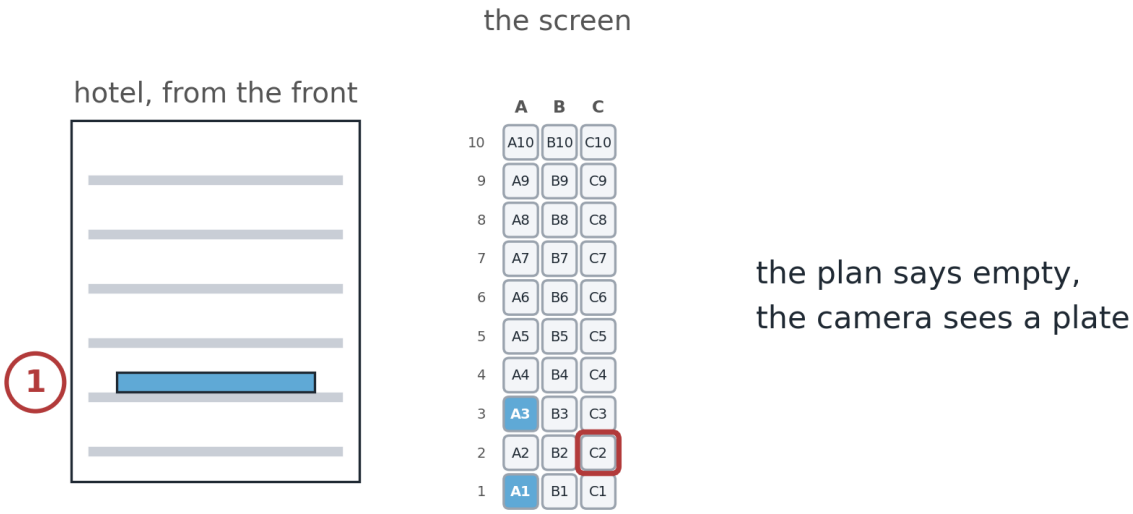
The arm holds still. Read the one line on screen. It is one of the three below.



1 one line on screen says what the arm saw. Read it, then the next three pages

6.1 A slot that looks taken

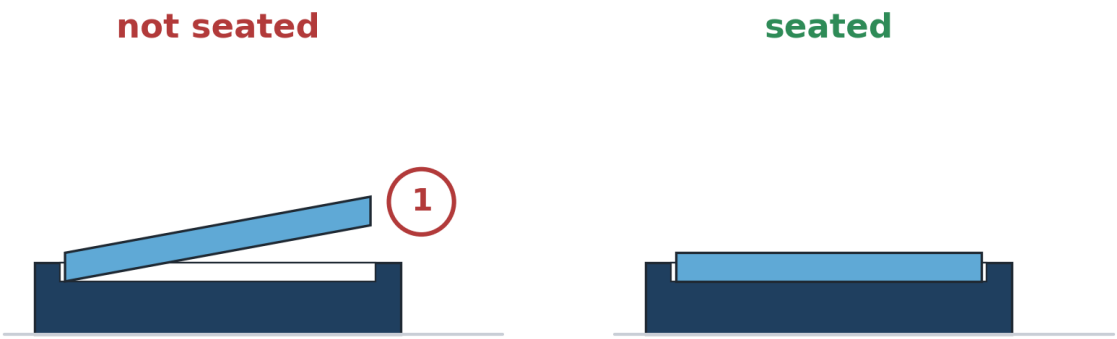
Take the plate out of that slot, or click the slot in Test. Then Start.



1 a plate someone left in a slot: take it out, or click that slot in Test. Then Start

6.2 A plate that does not look seated

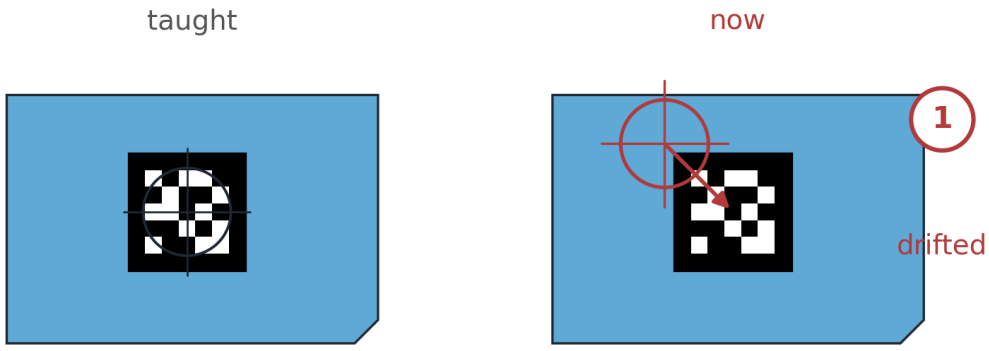
Seat the plate by hand, flat in the recess. Then Start.



1 on a corner, on the rim, or turned: seat it flat by hand. Then Start

6.3 A position that drifted

Something was bumped. Teach: find that row, Record again. Then Start.



1 something was bumped: Teach, that row, Record again. Then Start